CaltransGPT Using Retrieval-Augmented Large Language Models in Transportation Agencies

Rajnish Lekhi
Department Of Civil and Environmental Engineering
California State University
Fullerton, CA 92831
rlekhi@fullerton.edu

Akshat Desai
Department of Computer Science
California State University
Fullerton, CA 92831
akshat.desai.754@gmail.com

Rakeshkumar Mahto
Department of Electrical and Computer Engineering
California State University
Fullerton, CA 92831
ramahto@fullerton.edu

Deepak Sharma
Department Of Civil and Environmental Engineering
California State University
Fullerton, CA 92831
dsharma@fullerton.edu

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***Abstract*—Vast public transportation agencies, such as the California Department of Transportation (Caltrans), manage extensive repositories of technical documentation critical for project delivery, regulatory compliance, and operational consistency. However, traditional institutional knowledge retrieval systems are often inefficient, static, and time-consuming, making it difficult for employees to find accurate and context-specific information. Hence, in this paper, CaltransGPT, a domain-specific Large Language Model (LLM) assistant, is presented, designed to streamline knowledge management and optimize workflows. The performance of CaltransGPT is evaluated by submitting 15 multi-domain queries and measuring the response based on grounding, accuracy, conciseness, and source citation. The results indicated high fidelity in information retrieval, with minor gaps in citation references for select queries.**

Additionally, the paper identified various applications where CaltransGPT can be utilized to improve document management within Caltrans. Moreover, the challenges and precautions regarding deploying such an intelligent assistant throughout an organization are also presented. This work will pave the way for evaluating future domain-specific LLM applications that enhance knowledge management, improve workflow efficiency, and support decision-making processes in public sector organizations.

***Index Terms*—Large Language Models (LLMs), Workflow Optimization, Retrieval-Augmented Generation (RAG), Knowledge Management Systems**

I. Introduction

Due to growing demands for public infrastructure and a simultaneous increase in the scrutiny from regulators, it has become very challenging for public transportation agencies such as the California Department of Transportation (Caltrans) to operate effectively and ensure that various interdepartmental agencies are working in tandem [1]. According to the fact sheet published in 2023, Caltrans has close to 21,000 employees who are working on projects ranging from planning and project delivery to highway maintenance, environmental compliance, and procurement, with budgets from $100,000 to several million dollars. For a vast organization like Caltrans to operate efficiently and effectively deliver its mission, it typically disseminates vast volumes of technical and administrative documentation, that includes Highway Design Manual (HDM), Local Assistance Procedures Manual (LAPM), California Manual on Uniform Traffic Control Devices (CA MUTCD), and the Project Development Procedures Manual (PDPM), among many others. Hence, documentation management is the backbone of maintaining regulatory compliance and engineering consistency throughout the agency.

Despite significant digitization of all documents, content management and access to institutional knowledge are cumbersome and fragmented. For example, employees often must rely on tedious keyword searches through static PDFs, repetitive email inquiries, or informal knowledge passed between colleagues. This inefficient information retrieval process increases project delivery time, avoidable errors in documentation and approvals, inconsistent policy interpretation, and redundant administrative tasks. Such confusion has often led to unnecessary lawsuits that have cost millions of dollars in legal expenses and halted progress in project development. For instance, in Citizens for a Responsible Caltrans Decision vs. Department of Transportation, where Caltrans claimed an exemption from the full environmental review process under Streets & Highways Code 103 for the freeway interchange project in San Diego (where Interstate 5 meets State Route 56), the court found the exemption invalid and reinstated the lawsuit [2]. Similarly, in another instance, the California State Auditor in April 2001 found one of the Caltrans employees involved in conflicts of interest and resource misuse, highlighting deeper lapses in internal ethics and documentation procedures [3]. Hence, besides the digitalization of internal and external documents, there is a pressing need for intelligent knowledge management systems that can provide contextual, accurate, and timely access to information across departments.

Recently, due to breakthroughs in generative artificial intelligence (AI), especially Large Language Models (LLMs), such as OpenAI's GPT-4, Meta's LLaMA, and Google's

Gemini, have shown how these technologies can transform organizational management [4]. These technologies have the potential to enable rapid access to institutional knowledge, automate routine tasks, and support informed decision-making through natural language interfaces. As shown in [5], LLMs have been adopted for knowledge discovery and data mining in businesses, the pharmacy and healthcare industries, educational institutions, public policy, and many others. Similarly, Reitemeyer et al. presented an innovative solution for electronic court filing using ArchiMate, reducing the manual effort required for linking concepts and building system models [6]. In another instance, the Washington Metropolitan Area Transit Authority has developed and deployed a domain-specific LLM, MetRoBERTa, to take feedback from riders and improve their customer service experience [7]. These proven uses of LLMs in other large organizations like the Washington Metropolitan Area Transit Authority demonstrate that when trained using the agency-specific documentation and workflow, they can significantly enhance internal knowledge transfer, reduce response time, and support informed decision-making across departments.

This study presents the architecture of a domain-specific LLM assistant, CaltransGPT, trained using documents on the Caltrans website to support agency-wide knowledge management and workflow optimization. Additionally, this paper highlights the benefits and challenges associated with the large-scale adoption of LLMs by Caltrans and the precautions they can take to ensure responsible use, data security, and minimize misinformation. Hence, the goal of this paper is threefold: (1) to identify key opportunity areas within Caltrans operations where LLMs can provide measurable benefits, (2) to outline implementation challenges specific to public sector constraints such as governance, data privacy, and accuracy assurance, and (3) to offer a forward-looking perspective on how generative AI can help modernize public agency workflows. Given the scale of transportation agency operations, the proliferation of documentation, and the critical need for agility and institutional memory, this work contributes timely insights for public administrators, digital infrastructure planners, and AI practitioners seeking to align emerging technologies with public service goals.

II. System Architecture

This section describes the system architecture and deployment design of the smart knowledge assistant, CaltransGPT, to properly disseminate institutional knowledge across a vast organization like Caltrans. This intelligent assistant is built around Retrieval-Augmented Generation (RAG) to deliver context-aware answers from Caltrans' official documentation [8]. The architecture of CaltransGPT combines keyword-based and semantic search mechanisms for retrieving accurate information from a 15GB corpus of Caltrans technical documentation, as shown in Fig. 1. CaltransGPT's architecture is developed to respond to structured queries using official content from internal manuals, standards, and policy documents.

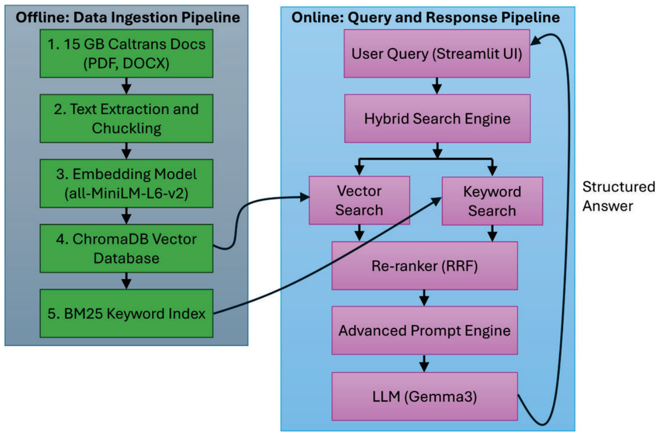

Fig. 1. System architecture of the RAG-based hybrid search framework. The architecture consists of an offline data ingestion pipeline and an online query–response pipeline.

A. System Design and Architecture

As shown in Fig. 1, the data pipeline begins with an offline data ingestion process, where the Caltrans document repository, which comprises engineering specifications, procedural guidelines, environmental standards, and other documents, is converted into raw text. The data is then semantically segmented into coherent chunks. This process of dividing lengthy technical documents into smaller, semantically meaningful units enables the retrieval of accurate and contextually relevant information during the search and facilitates effective response generation. Each chunk is then converted into a dense vector representation using the all-MiniLM-L6-v2 transformer, which are then indexed and stored in ChromaDB vector databases.

III. Hybrid Retrieval And Deployment

A. Hybrid Retrieval Mechanism

When a user submits a query, the system initiates a hybrid search process that consists of traditional keyword-based retrieval through BM25 and semantic vector-based similarity matching, as shown in Fig. 1. The results from both search techniques are merged through Reciprocal Rank Fusion (RRF). The RRF assigns a higher score if the documents appear higher in either of the search processes. This ensures that the most valuable results from both methods are returned without missing anything important. The top-ranked documents are then embedded with dynamically constructed prompts and passed on to the Gemma-3 LLM, which can generate more concise and grounded answers. This architecture has been validated for synthesizing context-specific information and returning well-structured responses based on officially sourced data.

B. User Access and Deployment

The proposed CaltransGPT-like smart assistant for effective institutional knowledge dissemination should be deployed in a dual-access configuration. One of the versions should be available to the general public through a web interface to

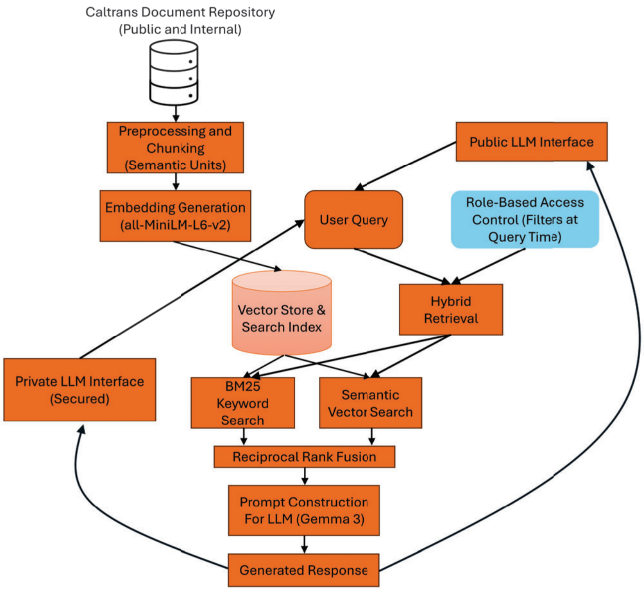

Fig. 2. The dual-access architecture of the proposed Caltrans Knowledge Assistant system.

provide easy access to officially published Caltrans documents and increase transparency in their governance. An advanced version with a secured internal interface should be made available to staff members to access confidential or restricted unpublished technical documentation and information. The architecture ensures differentiated access based on user roles while maintaining a unified backend system for scalability and maintainability.

In both cases, users submit their natural language query requests through the respective user interface (UI). Both requests are processed through the hybrid retrieval system shown in Fig. 2, which combines BM25 keyword-based search with semantic vector search using all-MiniLM-L6-v2 embeddings. A key feature of the architecture shown in Fig. 2, is that even when the public does not have access to restricted internal documents, the proposed architecture uses the same shared document repository while enforcing role-based access control. Though the underlying embedding store and retrieval engine index the entire corpus, including both public and confidential documents, document filtering is enforced dynamically at query time to ensure that public queries can only access those documents that are marked as public-level access. Top-ranked document retrieval is done through the RRF, which is later used to generate a context-aware prompt through the Gemma-3 LLM model. The generated information is returned to the appropriate user interface (internal or public), ensuring access to accurate and context-specific institutional knowledge.

Although the presented architecture looks promising, measuring whether the CaltransGPT presented in Fig. 1 can make knowledge retrieval more efficient and dynamic is essential. Hence, in the following section, we present an evaluation of the performance of this intelligent knowledge assistant through a preliminary testing protocol.

IV. Evaluation and Results

The proposed RAG-based knowledge assistant, CaltransGPT, offers an innovative solution for disseminating institutional knowledge and improving user experience. However, evaluating its performance is essential to assess the effectiveness, grounding, and reliability of its responses. As highlighted by Arslan et al. , rigorous performance assessment of generative AI applications is critical in ensuring trust, usability, and real-world impact, particularly in domain-specific deployments such as public sector knowledge systems [9].

A. Evaluation and Qualitative Criteria

To evaluate the performance of CaltransGPT, we created 15 domain-specific queries to conduct a functional evaluation of CaltransGPT, as shown in Table 1. These queries are carefully designed to ensure that both internal and external users can request from the CaltransGPT. The queries span six major operational areas: Highway Design Standards, Project Development Procedures, Construction and Resident Engineering, Traffic Control Devices, Procurement and Funding (LAPM), and Policy Interpretation.

Each of the queries is shown in the Table. I were submitted to CaltransGPT through the UI, and the generated responses were then subjected to a structured manual evaluation process, focusing on four key qualitative criteria: grounding, accuracy, conciseness, and source citation. Grounding refers to whether

TABLE I
Sample domain-specific queries covering six major Caltrans operational areas used for evaluating the assistant

Domain Area	**List of Queries**
Design Standards (Highway Design Manual – HDM)	Q1. What is the minimum lane width required for rural highways according to the 2023 HDM?
	Q2. How are sight distance requirements determined in horizontal curve design?
	Q3. What are the specifications for crash cushion design in the HDM?
Project Development Procedures (PDPM)	Q4. What are the major milestones in the PS&E (Plans, Specifications, and Estimates) process?
	Q5. Who is responsible for the environmental review under the PDPM?
	Q6. How is the Project Initiation Document (PID) structured and what must it include?
Construction & Resident Engineering (Construction Manual)	Q7. What are the duties of a Resident Engineer during construction inspection?
	Q8. How should a change order be submitted and approved according to Caltrans policy?
	Q9. What are the required forms for a Notice of Completion submission?
Traffic Control Devices (CA MUTCD)	Q10. What are the different types of channelizing devices defined in the CA MUTCD?
	Q11. How is sign retroreflectivity assessed and maintained?
	Q12. When are temporary traffic control plans required for maintenance work?
Procurement, Funding & LAPM (Local Assistance)	Q13. What documentation is required to request federal reimbursement under the LAPM?
	Q14. Which forms must be submitted for Right of Way certification?
Policy & Interpretation	Q15. What is the difference between a “standard” and a “guideline” in Caltrans documentation?

TABLE II
QUALITATIVE EVALUATION OF CALTRANSGPT RESPONSES ACROSS 15 DOMAIN-SPECIFIC QUERIES USING FOUR KEY CRITERIA: GROUNDEDNESS, ACCURACY, CONCISENESS, AND SOURCE CITATION

Query Numbers	Groundedness	Accuracy	Conciseness	Source Citation
Q1	(✓)	(✓)	(✓)	(✓)
Q2	(✓)	(✓)	(✓)	(✓)
Q3	(✓)	(✓)	(✓)	(✓)
Q4	(✓)	(✓)	(✓)	(✓)
Q5	(✓)	(✓)	(✓)	(✓)
Q6	(✓)	(✓)	(✓)	(✓)
Q7	(✓)	(✓)	(✓)	(✓)
Q8	(✓)	(✓)	(✓)	(✓)
Q9	(✓)	(✓)	(✓)	(✓)
Q10	(✓)	(✓)	(✓)	(✓)
Q11	(✓)	(✓)	(✓)	(✓)
Q12	(✓)	(✓)	(✓)	(✓)
Q13	(✓)	(✓)	(✓)	(✓)
Q14	(✓)	(✓)	(✓)	(✗)
Q15	(✓)	(✓)	(✗)	(✗)

the retrieved information is traced back to the official Caltrans document. Similarly, it is also essential to verify the accuracy of the generated information by the CaltransGPT; this is done to ensure that the assistant does not misinterpret, generalize, or distort technical content. Also, it ensures the users receive information that faithfully reflects the original Caltrans policies and standards. Besides grounding and accuracy of the information generated by CaltransGPT, the response should be concise and to the point, where the query can be directly addressed in a streamlined manner. Hence, conciseness as a metric is added to ensure the response is not overly verbose or includes tangential information. Finally, the CaltransGPT should also provide a verifiable reference to the user, such as a manual name, a specific section number, or a publication year. Each of the responses to the selected 15 queries will be either marked as meeting (✓) or not meeting (✗) in either of the four metrics.

B. Preliminary Results

The responses generated by the CaltransGPT assistant for the fifteen evaluation queries were assessed against four key qualitative metrics: groundedness, accuracy, conciseness, and source citation. The evaluation results are presented in Table II.

All fifteen responses generated by CaltransGPT were determined to be well-grounded, with the information in each response directly traceable to the official Caltrans documentation corpus. The assistant consistently retrieved and summarized relevant content from authoritative sources, thereby improving the reliability and interpretability of the generated responses.

An example response generated by CaltransGPT for Query 1 is summarized below:

Example Query: What is the minimum lane width required for rural highways according to the 2023 HDM?

Response Summary: The assistant identified that the standard minimum lane width for most highways is 12 feet. However, it also recognized documented exceptions for conventional state highways, where lane widths of 11 feet or 10.5 feet may be permitted depending on factors such as roadway classification, traffic volume, operational conditions, and geometric design constraints.

Source Reference: The response was grounded in the California Highway Design Manual (HDM), with references to the relevant sections describing lane width standards, allowable exceptions, and associated roadway design considerations.

Similarly, all the responses generated by the CaltransGPT were accurate, with no instances of misinterpretation or misinformation detected across the evaluated queries. Although the responses to queries 14 and 15 were grounded and accurate, the assistant could not provide a reference to the corresponding Caltrans manuals from which the information was retrieved. In conciseness, fourteen of the fifteen responses were succinct and directly addressed the user query without unnecessary elaboration. However, the response to Query 15, which was about differentiating between standards and guidelines, had additional explanatory content. Although the extended response was informative, it exceeded the scope of the original question and was considered verbose. For Queries 14 and 15, the CaltransGPT was able to provide accurate information; however, it failed to provide a reference to the exact section or subsection numbers of the corresponding Caltrans manual from which this information was retrieved. Hence, we considered the response for this query was classified as "not meeting" as shown in Table II.

The preliminary evaluation demonstrates the effectiveness of CaltransGPT in delivering accurate and contextually relevant information with references. However, further analysis is required to fully assess its performance. Understanding the broader implications of deploying such an intelligent assistant across a vast organization like Caltrans is also crucial. Hence, the following section discusses the potential application of CaltransGPT in other applications besides retrieving institutional information. Besides those applications, it is essential to recognize the challenges associated with AI technology and the precautions that should be taken before deploying such a system.

V. DISCUSSION

The preliminary evaluation of CaltransGPT demonstrates its potential as a transformative tool for institutional knowledge management within large public infrastructure agencies. CaltransGPT demonstrates its ability to generate accurate, well-grounded responses to queries across various operational domains, including highway design, construction management, policy compliance, and procurement workflows.

A. Application of CaltransGPT in Agency Workflows

Integrating CaltransGPT into day-to-day operations at the agency offers several opportunities to address long-standing document retrieval inefficiencies, support procedural compliance, and improve internal communication. One of the most immediate use cases is semantic search and retrieval from

Caltrans's vast repository of technical manuals and policy documents. Unlike traditional keyword-based systems, CaltransGPT enables staff and the general public to ask natural-language queries and receive concise, contextually relevant answers with precise document references. This reduces reliance on time-consuming manual searches or informal knowledge transfers.

Another area where CaltransGPT can assist is in document drafting and form completion. While completing routine tasks such as preparing environmental review narratives, drafting coordination memos, or completing procurement forms, employees must consult multiple documents to extract the relevant information. In such a scenario, CaltransGPT can serve as a one-stop information resource that they can use as a centralized information source. Generating draft content based on structured inputs and past document patterns allows human experts to focus on content validation rather than repetitive drafting. Similarly, CaltransGPT can assist in correctly classifying and tagging the document to ensure it is stored and organized properly in the database or content management system. Often, improper tagging or labeling leads to difficulties with document retrieval, duplication of effort, and inconsistencies in compliance reporting. The use of CaltransGPT here can significantly enhance operational efficiency. In cross-divisional coordination, CaltransGPT can summarize project status updates, extract action items from multi-stakeholder meetings, and provide real-time responses to policy interpretation questions. This capability can significantly reduce communication bottlenecks across planning, design, environmental, and procurement units.

Furthermore, CaltransGPT can be installed or accessed on a mobile device, enabling it to support field operations and maintenance reporting. Therefore, it can aid in transcribing and structuring field inspection notes, improving field personnel's data integrity and turnaround times. Additionally, its role in training, onboarding, and institutional knowledge transfer is critical in mitigating the loss of tacit knowledge due to staff turnover and retirements.

B. Challenges and Implementation Considerations

Despite these promising applications, several challenges must be addressed to ensure the responsible and effective deployment of CaltransGPT. The assistant's performance depends on the quality of the input documents uploaded. Often, the documents are in unstructured PDF or image format, making it difficult for the model to extract accurate, relevant information. Furthermore, documents often use inconsistent terminology, leading CaltransGPT to respond incorrectly. Hence, developing a well-curated, semantically segmented knowledge corpus is vital to improving information retrieval accuracy. Hallucination is a significant challenge associated with LLMs [10]–[12], in which models may generate incorrect or fabricated information despite appearing confident. Such incorrect responses can severely affect regulatory compliance and engineering standards. Hence, it is essential to adopt strict grounding mechanisms with inline citation requirements to obtain a trustworthy and reliable response.

The dual-access architecture plays a crucial role in improving public and private access to Caltrans information; however, in such a scenario, robust data governance and security measures are essential. Hence, the dual-access architecture should strictly comply with state and federal privacy mandates by implementing RBAC and protecting sensitive information from unauthorized access. Therefore, it is necessary to have a robust architectural design integrating identity management, secure hosting environments, and immutable interaction logs.

Additionally, using LLMs in CaltransGPT, along with vector search and real-time response expectations, makes the system computationally and resource-intensive. Hence, deploying such a system, accessible throughout the organization and on mobile devices, requires a scalable, secure cloud-based infrastructure. All these infrastructures require additional operational costs, which Caltrans needs to account for to make the program sustainable.

Finally, organizational readiness and change management are critical factors. Successful adoption depends on user education, trust-building, and demonstrating clear value in day-to-day workflows. Concerns about AI replacing human judgment or bypassing established review processes must be proactively addressed through structured training and pilot programs.

C. Future Work and Enhancement Pathways

Though the CaltransGPT provided accurate, concise responses with reference to users' queries from various domains in the evaluation, further assessment is still required. Hence, a pilot study is essential for the Caltrans staff and select public users to interact with CaltransGPT. They then further share their experience in assessing usability, accuracy of responses, and overall user satisfaction to refine the assistant's capabilities before large-scale deployment. Additionally, in this work, we only proposed the dual-access architecture of the CaltransGPT. The robust evaluation of such architecture also needs to be done to ensure that, before it is deployed, its scalability, reliability, and effectiveness in the real world are ensured across various operational scenarios.

Furthermore, many organizations in the United States have access to an AI platform such as Microsoft Co-Pilot, which has built-in features to ensure organizational data privacy and compliance. Future work will also evaluate CaltransGPT in privacy-preserving environments such as Microsoft Copilot, considering deployment cost, time, and operational effort compared to existing enterprise AI solutions.

In future work, we will report further improvements in the CaltransGPT. Firstly, we will develop an automated architecture that allows information to be added without manual intervention, ensuring that Caltrans policies remain up to date in the assistant. Though the current version of CaltransGPT can provide the source of the information it retrieves, we will continue to improve it further. In addition, we will update the CaltransGPT to understand other types of information, such as data tables and engineering drawings, not just text. Moreover, we will be adding a feedback mechanism to ensure users can point out mistakes or suggest better answers, allowing the assistant to get smarter over time. All these improvements will ensure that CaltransGPT becomes a trusted and helpful tool supporting Caltrans employees in their daily work and helps the agency serve the public more efficiently.

VI. Conclusion

The initial evaluation included 15 queries from different domains, such as highway design, project development, construction, and procurement. CaltransGPT was able to generate grounded, accurate, and contextual responses with references, highlighting its capabilities. Only two queries showed minor limitations, which were general in nature. In those cases, CaltransGPT generated responses that marginally missed the concise and source citation requirements.

Beyond technical performance, deploying CaltransGPT within a public agency context raises broader considerations regarding data governance, system integration, user trust, and ethical AI use. This paper highlighted different applications where the CaltransGPT can be crucial in improving the organization's overall efficiency. This paper also presented key implementation challenges, including hallucination risks, compliance requirements, infrastructure constraints, and cultural readiness. Moreover, the paper presents the proposed dual-access architecture as a roadmap for deploying CaltransGPT across Caltrans workflows.

Future work discusses further evaluation of the CaltransGPT to measure the effectiveness of the trained intelligent assistant. The paper also highlighted and identified other advancements to improve the capabilities of CaltransGPT. As generative AI becomes more intelligent and capable, the metrics presented in this work can be utilized in different organizations to test the performance of organization-specific LLMs for better institutional knowledge management, policy compliance, and operational decision support.

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